

PURNA MUMMANI

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Education

University of North Carolina at Charlotte GPA: 3.8/4

Jan 2024 – Dec 2025

Master of Science in Computer Science (Concentration in Data Science)

Charlotte, NC

Vellore Institute of Technology, India

Aug 2019 – July 2023

Bachelor of Technology in Computer Science

India

Experience

UNC Cahrlotte: OneIT

Aug 2024 – May 2025

Student IT Support Technician

Charlotte, NC

- * Delivered frontline IT support by resolving **15–20 weekly technical issues** via the **Mercury ticketing system**, troubleshooting Wi-Fi, hardware, and software for Windows/macOS devices; facilitated campus-wide move-in technology setups by onboarding resident devices and coordinating with senior IT staff to escalate complex incidents while ensuring clear end-user communication and 100% incident resolution.

Cognizant

Jan 2023 – Dec 2023

Software Engineer

Hyderabad, India

- * Optimized **React.js** UI responsiveness and component reusability to **speed up page loads by 28%**, while architecting **Next.js SSR/REST APIs** to drive a **22% increase** in organic traffic.
- * Migrated legacy features to **Spring Boot/Docker** microservices to reduce service errors by **24%** and integrated **Redis** caching to decrease API response times by **18%** for high-concurrency handling.
- * Enforced robust security protocols, including **JWT, RBAC, and AES encryption**, to protect sensitive data for **50+ internal users** while maintaining strict organizational and data compliance standards.

Cognizant

July 2022 – Dec 2022

Intern

Hyderabad, India

- * Developed backend features using **Java/Spring Boot** and executed **100+ tests** to cut bugs by **18%** while reducing requirement gaps by **12%** via **Postman API testing**.
- * Increased code coverage by **25%** using **JUnit/TestNG** and managed **CI/CD pipelines** to ensure seamless production releases with zero downtime and optimized reliability.

Technical Skills

Programming: Java, Python, C# , C, C++, HTML/CSS, JavaScript, SQL

Frameworks/Softwares: Angular, .NET Core,ASP.NET, ReactJS, NodeJS, Spring Boot, W3af,Metasploit, Jira, WordPress, Apache spark, Hadoop, Kafka

Database & Cloud: MySQL, PostgreSQL, NoSQL (MongoDB), Entity Framework, MS Azure, AWS, Docker, Kubernetes, CloudFormation, API Gateway

Developer Tools: VS,Eclipse, Virtual Box, VM ware, Git, MSSQL, Swagger, Figma, Jupyter, Google Colab, Cisco, Tableau, PowerBI

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV

other Tools: Data Bricks, Social Engineering Toolkit, Burp Suite, Nmap, Sqlmap, Wireshark, SNORT, SQL workbench

OS: Windows, MAC, Linux, Kali Linux, Ubuntu

Projects

Netflix-GPT | *React, OpenAI, Firebase, TailwindCSS*

Nov 2025

- Engineered a high-performance movie streaming platform that lowered bounce rates through seamless navigation and integrated the TMDB API to boost user engagement by 35% with real-time content; further deployed **OpenAI** for AI-powered search, enhancing recommendations and accuracy, while optimized system performance to decrease load times by 25% through a mobile-responsive design.

Hybrid Transformer-CNN for Dense Video Analysis | *PyTorch,Python,Transformers,OpenCV*

Sep 2025

- Designed and implemented a full-scale MS-TCT (ConvTransformer) in PyTorch, utilizing a 4-stage temporal encoder to capture atomic and composite actions in untrimmed videos, achieving a **+9.8% mAP boost** over I3D baselines.

- Customized a Gaussian Heat-map branch and Temporal Scale Mixer to unify multi-resolution features, optimizing the system to use **85% fewer FLOPs** while maintaining real-time frame-level detection at an **8 FPS sampling rate**.

Career War Room: AI Resume Insight & Job Matcher | *Python, Ollama, NLP, Streamlit* **May 2025**

- Developed and configured a privacy-focused intelligence platform by migrating from cloud APIs to Local **LLMs (Ollama)**, utilizing SentenceTransformers for semantic similarity to rank job listings with 100% on-device data security.
- Integrated SerpAPI and Indeed RSS for real-time data retrieval and built a full-stack Streamlit dashboard that automates skill-gap analysis and profile-to-job alignment via an asynchronous **FastAPI** backend.

Doctor-On-Call: Healthcare Management System | *JavaScript, MongoDB, Express, React, Node.js* **Aug 2024**

- Built a multi-tiered healthcare portal featuring secure **JWT-based authentication**, multi-role authorization, and an automated doctor-onboarding pipeline and real-time status notifications, deployed on **AWS** for high availability. Built an optimized scheduling engine to handle concurrent appointment bookings while maintaining atomic data consistency across **MongoDB** clusters.
- Improved system reliability by introducing automated testing and regression checks, reducing post-release logic issues by ~40% while securing sensitive patient data and ensure high reliability during peak traffic.

US Aviation Intelligence: Market & Fare Dynamics Analysis | *Tableau, Dashboard, Visual Analytics* **May 2024**

- Built a Tableau Story for US aviation trends analyzing 10,000+ flight records, utilizing K-means clustering and geospatial mapping to identify carrier dominance and uncover pricing inefficiencies.

Achievements & Certifications

Awarded for Technical Excellence at Engineering Clinics Expo 2021 for developing a real-time, Arduino-based robotic hand system from a pool of 250+ competing teams. Microsoft Certified: Azure Fundamentals (AZ-900), AWS Academy Graduate, Emotion AI Certification : Facial Key-points Detection.